FORECASTING KENYAN 91 DAY T-BILL RATES

GROUP 6
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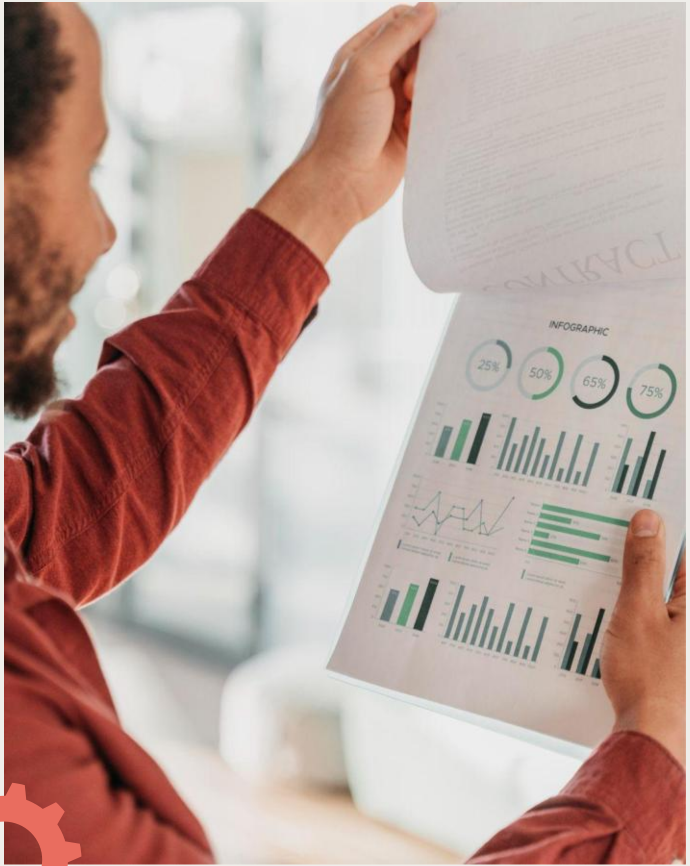
Introduction

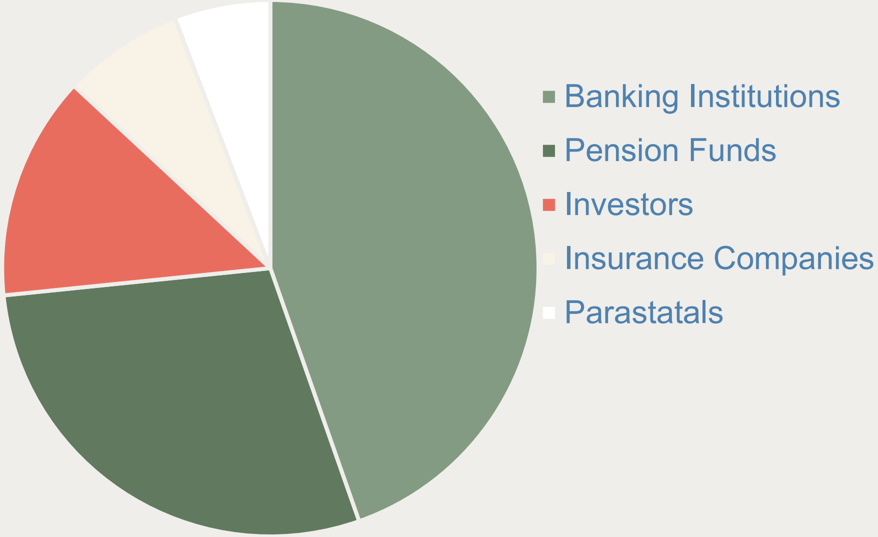
- KEY INVESTMENTS SIGNALLING SHORT TERM INTEREST RATES.
- DEPEND ON ECONOMIC FACTORS, MONETARY AND FISCAL POLICIES.
- HIGH TRANSACTION COSTS. (e.g Bloomberg)



Business problem

Since 1969, improvements made by Dhow CDS portal & app.

Domestic debt composition:



Problem statement

Bridge the gap by building a machine learning model to predict the 91-day T-Bill rate using public economic data.

Goal:
Provide accurate, transparent, and easy-to-use forecasts, empowering all market participants—not just financial elites—to make better investment decisions.

Data Used

91-day T-bill rate

Inflation

Public debt

USD Exchange rate

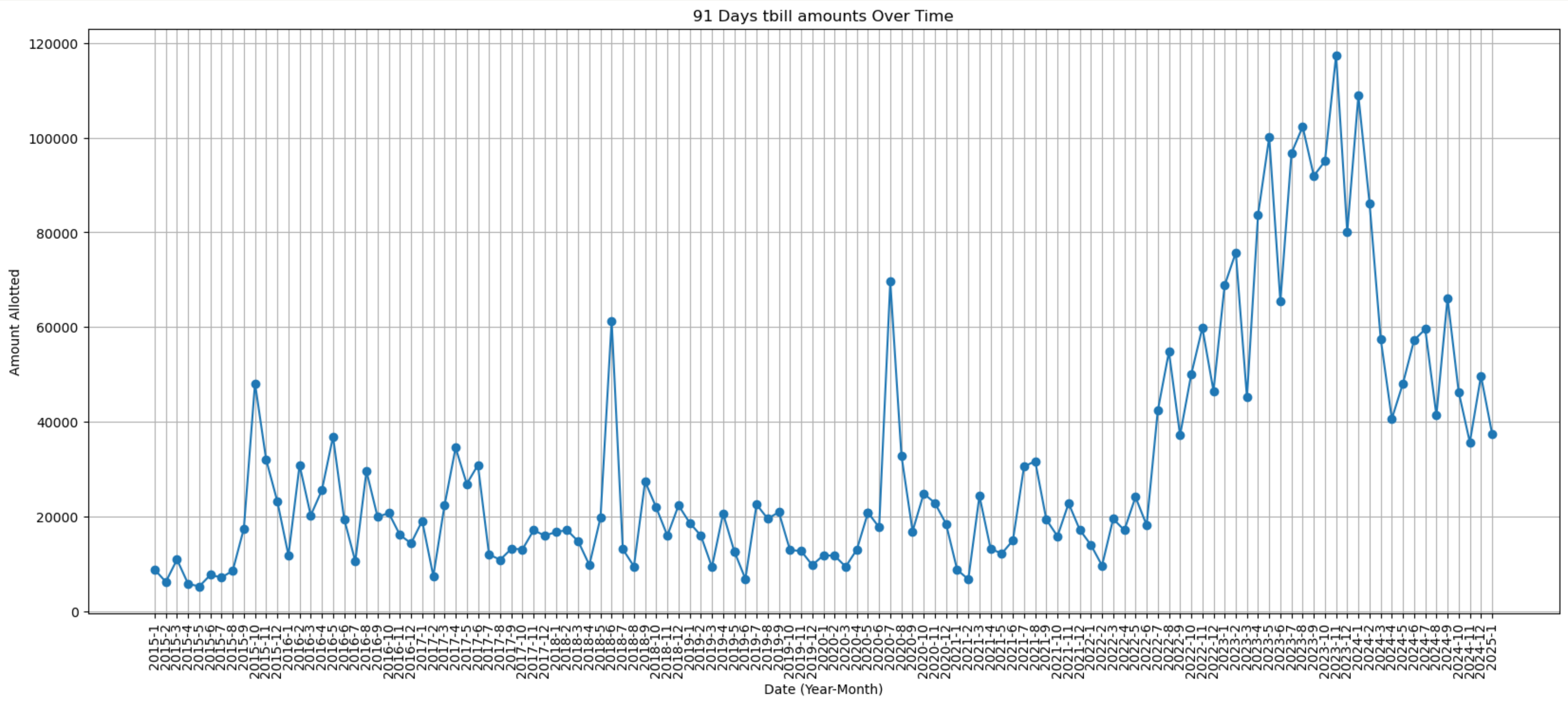
Amounts allotted & redeemed

Central Bank Rate

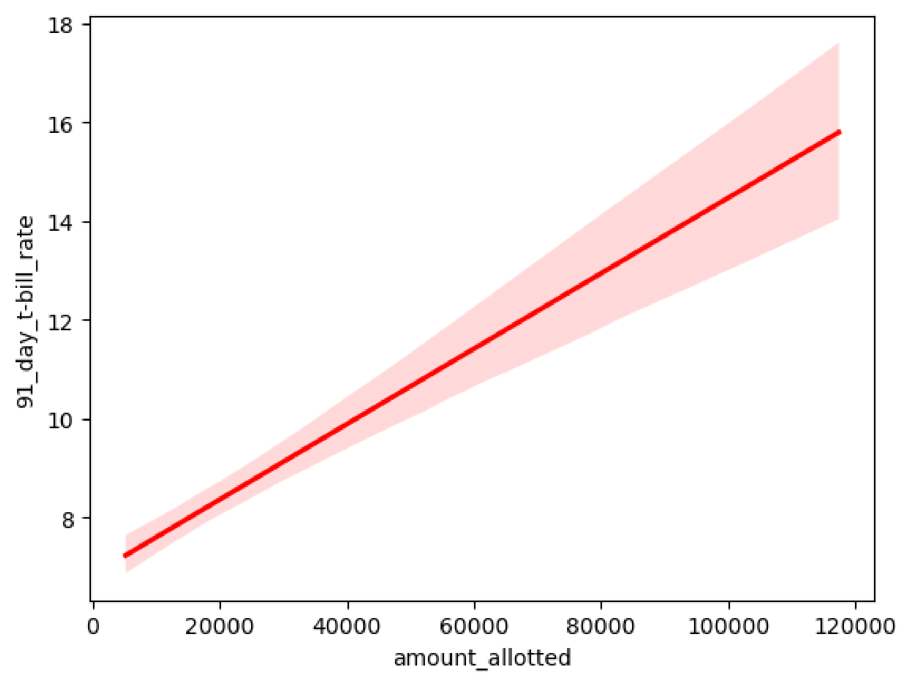
Annual GDP

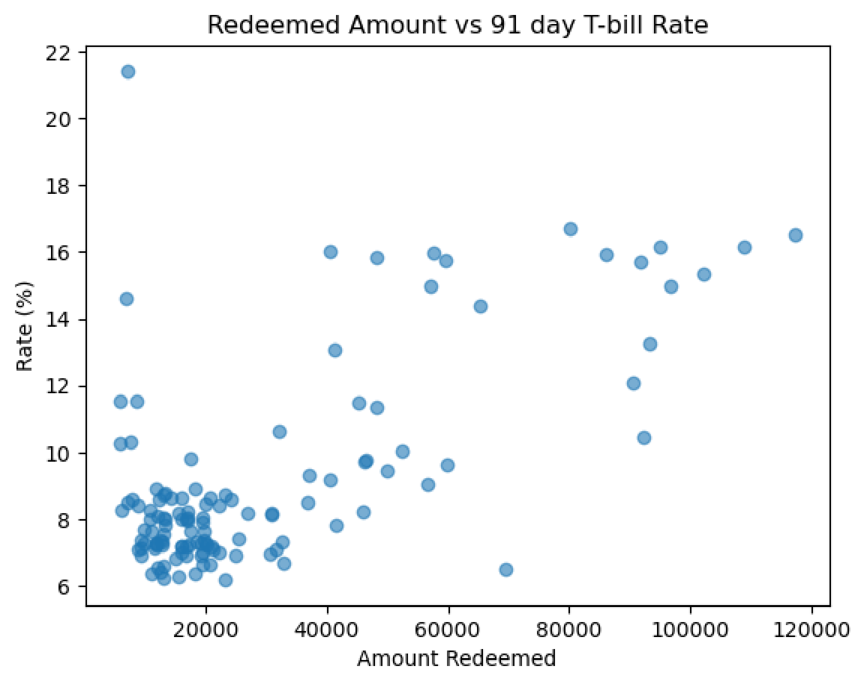
Sources: CBK, KNBS, WORLD BANK

91 Day T-bill rates over time

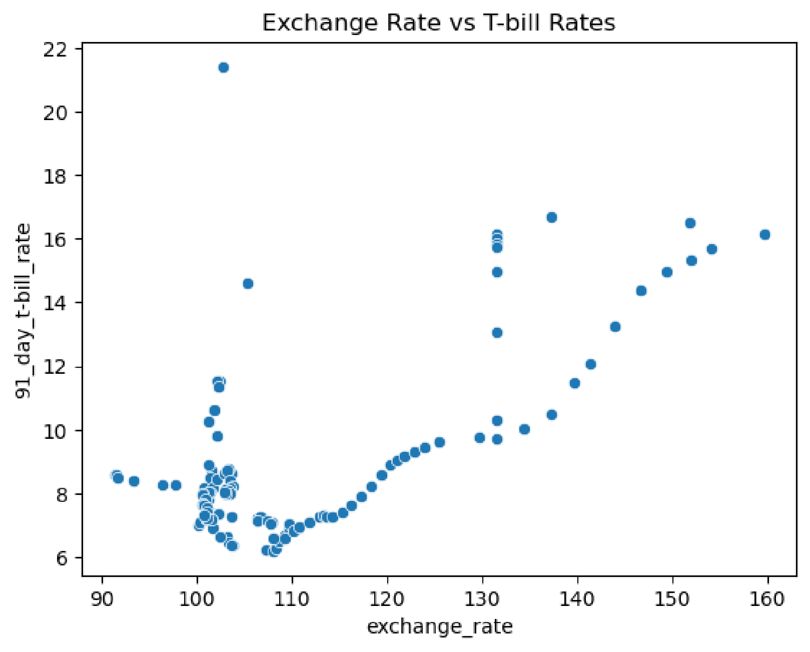


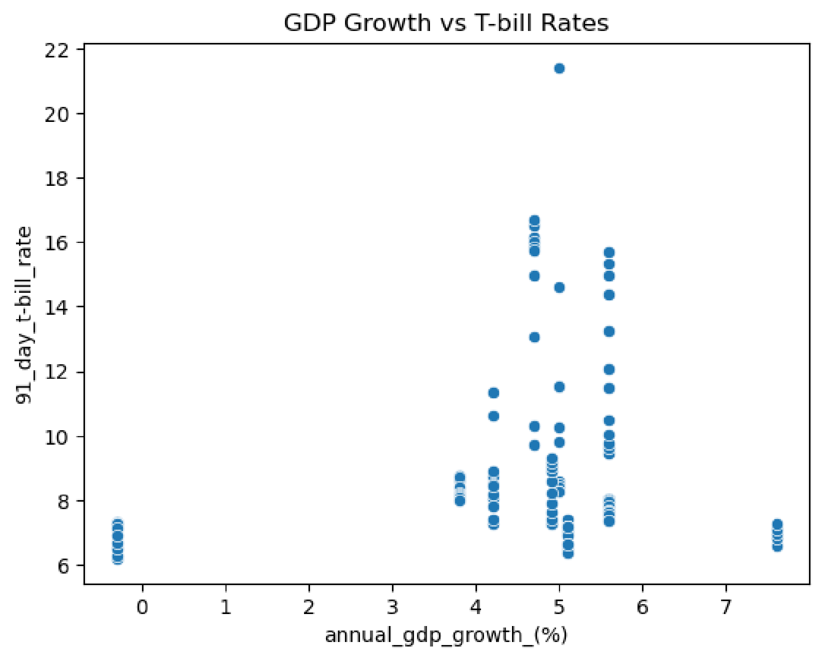
Two ideas


91_day_t-bill_rate
amount_allotted
18
16
14
12
10
8
0
20000
40000
60000
80000
100000
120000


Redeemed Amount vs 91 day T-bill Rate
Rate (%)
Amount Redeemed
22
20
18
16
14
12
10
8
6
20000
40000
60000
80000
100000
120000

Exchange rate and GDP growth vs T-bill rates





Models Used

Linear Regression	R-squared of 87.5%. Our regression explains the Y to an accuracy of 0.875.
ARIMA / GARCH	ARIMA was less favourable but GARCH still needed work.
Random Forest	RMSE (5.2854). Poor model in the test phase.
Light GBM	RMSE (0.3140). Chosen for deployment.

CHALLENGES

limited Data Scope: The model relies only on publicly available macroeconomic data, which may miss market sentiment, policy changes, or global shocks.

Feature Selection Bias: Important predictors could be omitted or underrepresented, affecting accuracy in unusual market conditions.

No Real-Time Updates: The system may not reflect the latest market events if data pipelines are not fully automated or updated frequently.

RECOMMENDATIONS

Regular Model Updates: Retrain the model periodically to account for changing economic conditions and prevent model drift.

Expand Data Sources: Incorporate additional variables such as market sentiment, global financial indicators, and policy announcements for improved accuracy.

Real-Time Data Pipeline: Automate data collection and model retraining to ensure forecasts reflect the latest market events.

Model Interpretability: Develop tools or dashboards that explain model predictions to non-technical users.

Scenario Analysis: Integrate stress-testing and scenario forecasting to help users understand model behavior under extreme market conditions.

CONCLUSION

LightGBM demonstrates strong predictive accuracy for the 91-day Kenyan Treasury Bill rate, with a low RMSE of 0.3140. This error is minor compared to the typical rate range, making the model suitable for practical financial decision-making. The system successfully integrates multiple macroeconomic indicators, automates data processing, and provides accessible forecasts for a wider range of market participants.

"An investment in knowledge pays the best interest."

—Benjamin Franklin

Thanks!

Do you have any questions?